

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Node.js + TypeScript Engineer with 5+ years designing REST API platforms, third-party integrations (Web3/DeFi, Morningstar, OpenAI, ffmpeg), and distributed services. Production-deep AWS migration experience (S3/EC2/EBS → GCS, \$5K/mo savings). Architected microservices with Apache Kafka + gRPC + NATS for multi-tenant case management. Mentored a 4-person engineering team. Available 4–6h overlap with US/Canada timezones from EAT.

CORE COMPETENCIES

Node.js + TypeScript

REST API Design

AWS (S3/EC2/EBS)

Microservices Architecture

Third-party API Integrations

Production Debugging

Mentorship

Async Remote Collaboration

WORK EXPERIENCE

Software Engineer

Sep 2022 – Present

MTailor (US Remote)

- **Real-time Pub/Sub two-way sync** between MongoDB and Firestore using **Node.js** and Google Cloud Pub/Sub for message processing — kept production traffic running during full backend migration.
- Led the **zero-downtime migration** of 20+ applications from Parse/MongoDB to Firebase/GCP, reporting directly to the CTO.
- Migrated all services off AWS, saving the company **\$5,000/month** in infrastructure cost; moved file storage from S3 to GCS via Python tooling.
- Built a 3D visualisation feature using video overlay + ffmpeg, increasing buyer conversion.
- Implemented Express Shipping pipeline, generating **+\$40 revenue per order**.
- Migrated a Webflow website from Parse to Firebase backends.

Node.js · Firebase · GCP Pub/Sub · AWS S3/EC2/EBS · Python · ffmpeg · React · Webflow · JavaScript

Software Engineer / Tech Lead

Jan 2020 – Jul 2021

CodeBits (Uganda Remote)

- **Architected a microservices backend** for FIDA Uganda using **Apache Kafka, NATS Streaming, gRPC, Docker, and Kubernetes** — independent service deploys for multi-tenant case management.
- Led a team of 4 developers; ran code review and architecture decisions across legal-tech systems for NGOs.
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC.
- Built a paralegal database for LASPNET Uganda + cross-platform mobile app (React Native + Expo) with Firebase push notifications.

Node.js · Apache Kafka · NATS Streaming · gRPC · Docker · Kubernetes · Firebase · React Native · USSD

Software Engineer

Jan 2022 – Jul 2022

Mind2matter.co (US Remote)

- Built backends for DeFi applications using **Web3 and Node.js**, integrating on-chain protocols and custom smart contracts.
- Delivered React UIs from Figma designs under tight client timelines, reporting directly to the CTO.

Node.js · Web3 · Firebase · React · Tailwind

Software Engineer

Aug 2021 – Jan 2022

Dr Wealth (Singapore Remote)

- Optimised a PWA delivering real-time stock prices from a Firebase backend; queried over **2 million Firestore records** via ad-hoc batch jobs to keep prices current from Morningstar APIs.
- Extended backend APIs using Firebase Cloud Functions and **Express.js** on Heroku.
- Shipped responsive UI pages with React and Tailwind.

Node.js · Express · Firebase · React · Tailwind

- Built migration tooling reading from MongoDB and writing to Firestore after complex processing logic.
- Wrote engineering documentation for the Firebase Node.js + Python SDKs adopted across the codebase.
- Trained the Ops team on the new Firebase Dashboard and backup procedures.

Node.js · Firebase · AWS S3/EC2/EBS · GCP · JavaScript · Python

PROJECTS

Healthcare WhatsApp Chatbot AI · Integrations

Integrated OpenAI + Google Vision for prescription OCR via WhatsApp Business API.

Node.js · OpenAI · Google Vision · WhatsApp Business API · Firebase

Real-time Pub/Sub Sync Pipeline Distributed Systems

Live MongoDB ↔ Firestore sync supporting parallel production traffic during a zero-downtime migration of 20+ apps.

Node.js · Google Pub/Sub · MongoDB · Firestore

FIDA Uganda Microservices Architecture

Multi-tenant case management for legal-aid NGOs with Kafka topics, NATS Streaming, and gRPC contracts on Kubernetes.

Node.js · Apache Kafka · NATS Streaming · gRPC · Docker · Kubernetes

EDUCATION

B.Sc. Computer Engineering · Makerere University

Aug 2018 – Sep 2022

Kampala, Uganda

LANGUAGES

Languages

English · Kiswahili

SKILLS

Proficient: Node.js, TypeScript, Express, Firebase, GCP, Google Pub/Sub, Apache Kafka, gRPC, NATS Streaming, Docker, Kubernetes, MongoDB, Firestore, SQL, REST APIs, Git, Linux, Python, React

Intermediate: AWS (S3/EC2/EBS), Go, Rust, React Native, Web3